CYB102 Project 4

(🔗 Instructions Page)

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Reflection (Required)

> 🤔 **Reflection Question #1:** If I had to **explain "what is a proxy server" in 3 emojis,** they would be...
> (Feel free to put other comments about your experience in this unit here, too!)

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> 🧠**Reflection Question #2:** What are some different types of DoS/DDoS attacks?

Some different types of Dos and DDos attacks are:

- Volume-Based Attacks: (UDP Flood, ICMP [Ping] Flood, and SYN Flood)
- Protocol Attacks: (Smurf Attack, Ping of Death, and Teardrop Attack)
- Application Layer Attacks: (HTTP Flood, Slowloris, and GET/POST Flood)
- Fragmentation Attacks: (Fragmentation Overlapping Attack, and Teardrop Attack)

> 📣 **Shoutouts:** Share appreciation for anyone who helped you out with this project or made your day a little better!

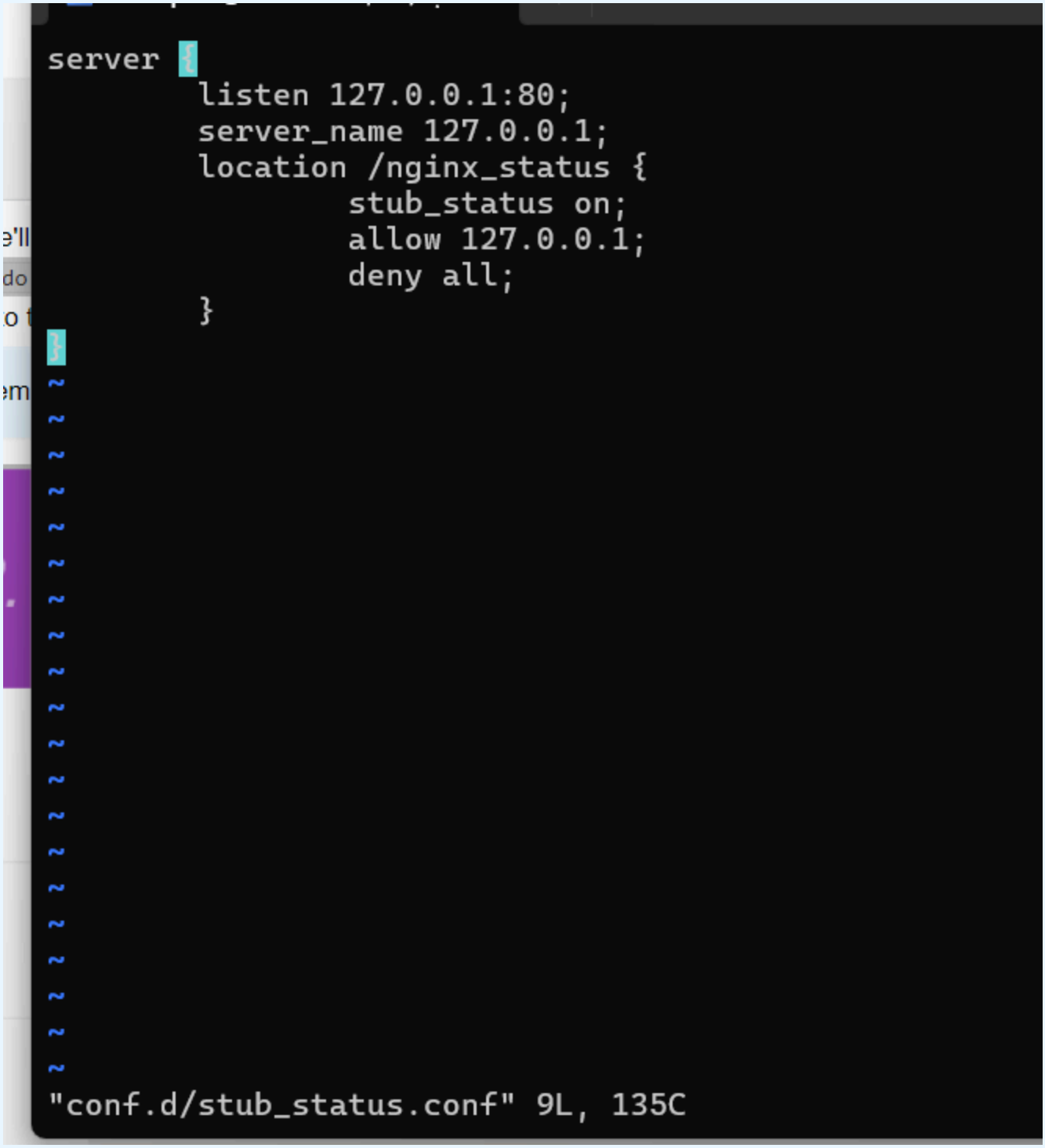
Big shoutout to Group #17!

Required Challenges (Required)

> **Item #1:** A screenshot of your `/etc/nginx/conf.d/default.conf` file with your DoS mitigation rules implemented:

[Insert Screenshot Here]

Screenshot of terminal with DoS mitigation Implemented: ** (Here I have my Dos Mitigation rules) **UPDATED for full credit grade



**SCREENSHOT BELOW IS THE FILE `/etc/nginx/conf.d/default.conf **`

```
root@lab000000:/etc/nginx/conf.d# cat default.conf
server {
    listen       80;
    server_name  localhost;

    #access_log  /var/log/nginx/host.access.log  main;

    location / {
        root   /usr/share/nginx/html;
        index  index.html index.htm;
    }

    #error_page  404              /404.html;

    # redirect server error pages to the static page /50x.html
    #
    error_page   500 502 503 504  /50x.html;
    location = /50x.html {
        root   /usr/share/nginx/html;
    }

    # proxy the PHP scripts to Apache listening on 127.0.0.1:80
    #
    #location ~ \.php$ {
    #    proxy_pass   http://127.0.0.1;
    #}

    # pass the PHP scripts to FastCGI server listening on 127.0.0.1:9000
    #
    #location ~ \.php$ {
    #    root           html;
    #    fastcgi_pass   127.0.0.1:9000;
    #    fastcgi_index  index.php;
```

```
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    listen       80;
    server_name  localhost;

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    location / {
        root   /usr/share/nginx/html;
        index  index.html index.htm;
    }

    #error_page  404              /404.html;

    # redirect server error pages to the static page /50x.html
    #
    error_page   500 502 503 504  /50x.html;
    location = /50x.html {
        root   /usr/share/nginx/html;
    }

    # proxy the PHP scripts to Apache listening on 127.0.0.1:80
    #
    #location ~ \.php$ {
    #    proxy_pass   http://127.0.0.1;
    #}

    # pass the PHP scripts to FastCGI server listening on 127.0.0.1:9000
    #
    #location ~ \.php$ {
    #    root           html;
    #    fastcgi_pass   127.0.0.1:9000;
    #    fastcgi_index  index.php;
    #    fastcgi_param  SCRIPT_FILENAME  /scripts$fastcgi_script_name;
    #    include        fastcgi_params;
    #}

    # deny access to .htaccess files, if Apache's document root
    # concurs with nginx's one
    #
    #location ~ /\.ht {
```

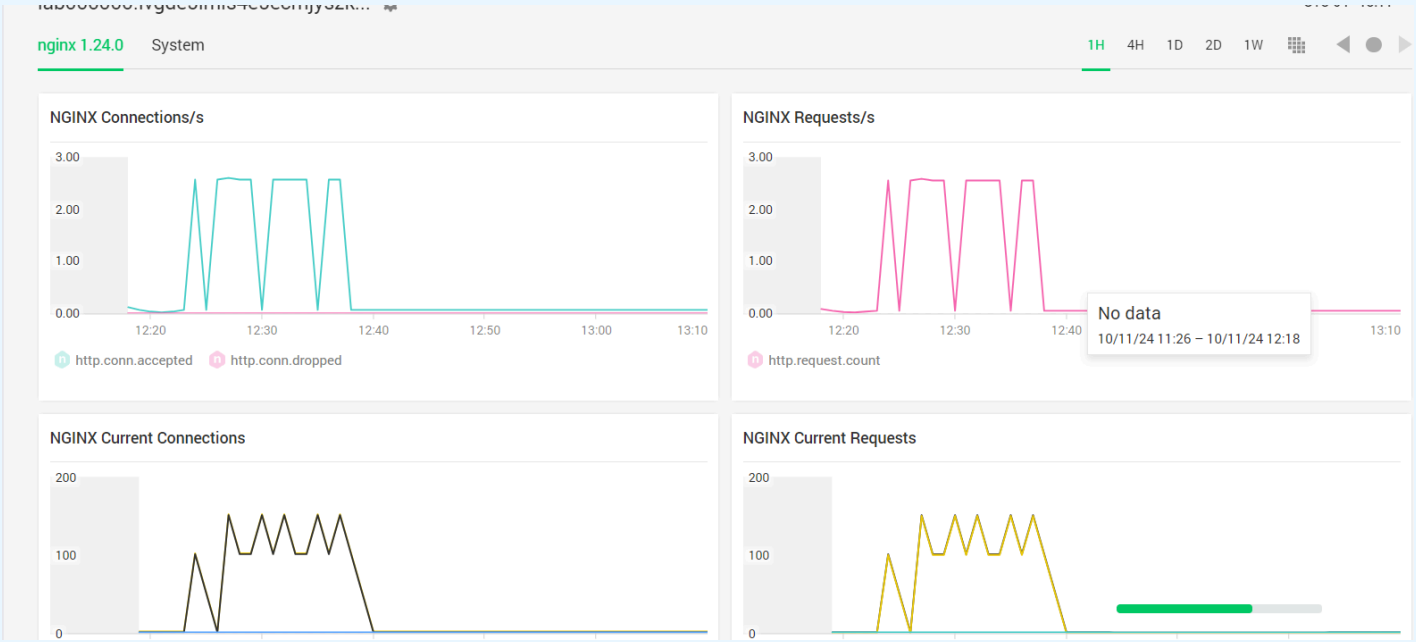
```
    #
    #location ~ /\.ht {
    #    deny  all;
    #}
}
```

Item #2: A detailed explanation (two sentences minimum) of how you know that your DoS mitigation rules are working:

My Dos mitigation rules are working because in Nginx Amplify, I can clearly see it is capturing attack requests/connections in my graphs. I know it works too because when I look in wireshark it only displays traffic under the TCP protocol for our server (127.0.0.1). However, when we compare the traffic captured for (127.0.0.53) we notice the traffic captured was under DNS proving there were no mitigation rules. We know the rules work too, because no

information was able to be revealed from our server after running our slowloris attack.

Screenshot (Optional):



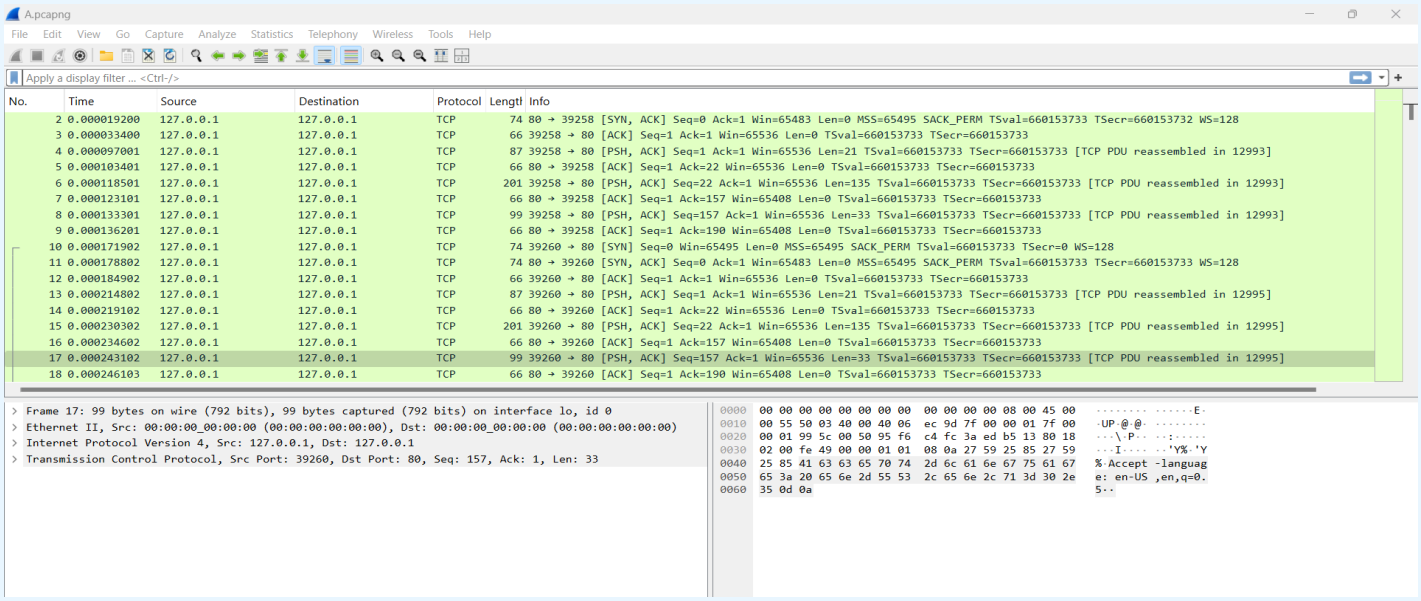

```
server {
        listen 127.0.0.1:80;
        server_name 127.0.0.1;
        location /nginx_status {
                stub_status on;
                allow 127.0.0.1;
                deny all;
        }
}
```

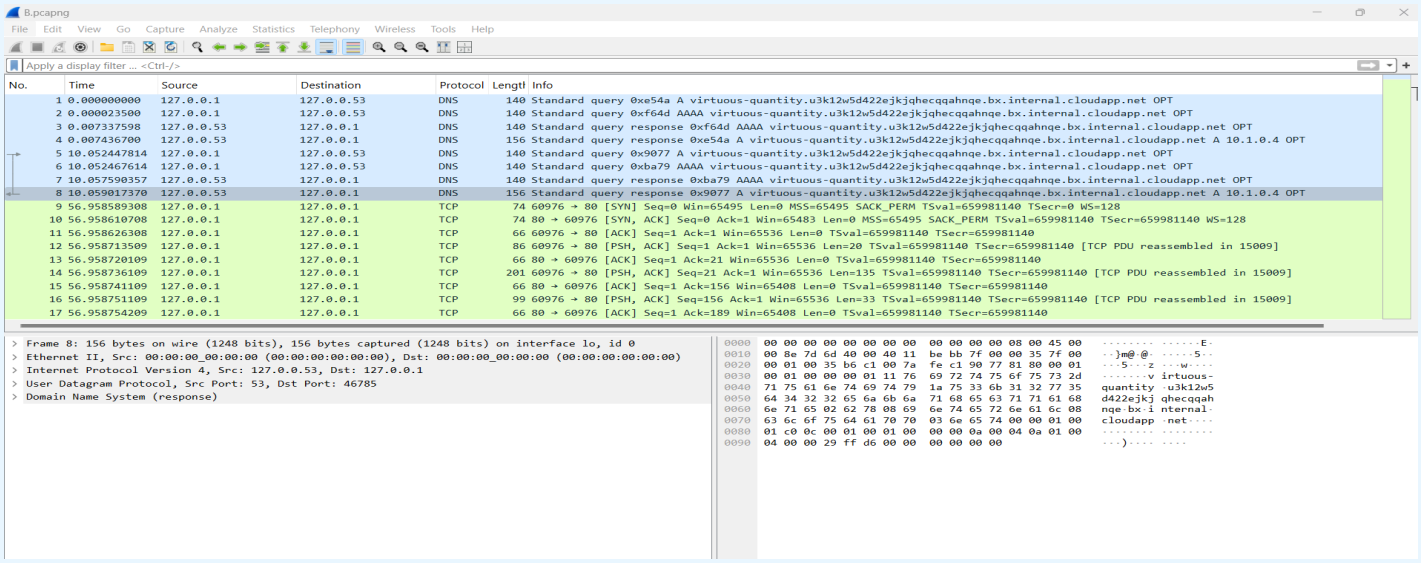
```
root@lab000000:/etc/nginx/conf.d# slowloris 127.0.0.1
[11-10-2024 12:12:51] Attacking 127.0.0.1 with 150 sockets.
[11-10-2024 12:12:51] Creating sockets...
[11-10-2024 12:12:51] Sending keep-alive headers...
[11-10-2024 12:12:51] Socket count: 150
```

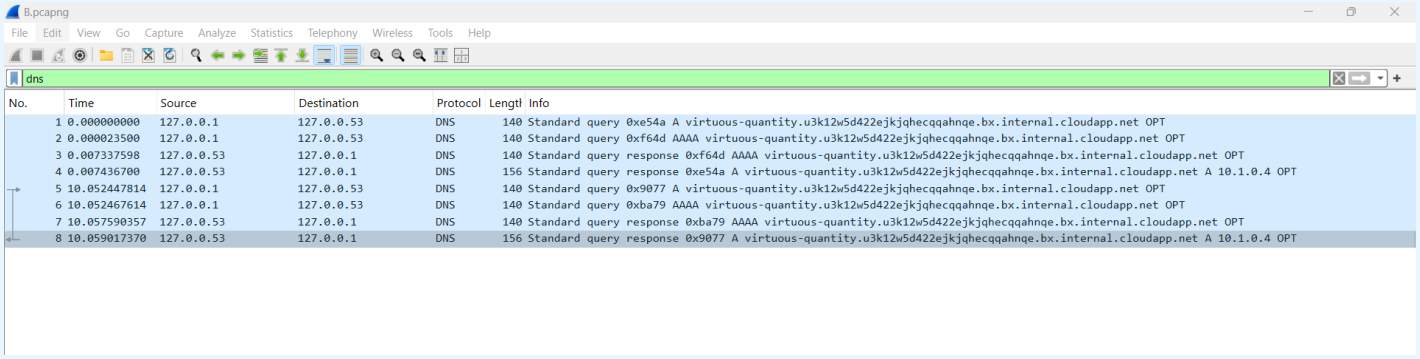
Item #3: A detailed explanation of how you know which `.pcap` file is from the vulnerable server, and which is from the server with DoS mitigation set up:

I know that the **B.pcap** file is from the vulnerable server coming from **(127.0.0.53)** and the **A.pcap** file is from the server with Dos Mitigation set up coming from **(127.0.0.1)** We know this based on a number of facts:

1. Looking at the A.Pcap file it only displays traffic coming to and from 127.0.0.1 with the protocol TCP and no real information was able to be attacked successfully.
2. Looking at the B.pcap file it displays traffic from 127.0.0.53 going to and coming back to 127.0.0.1, however in this file we notice the DNS protocol being revealed, exposing urls and the ability to perform Dos attacks.
3. We also know that when we ran the slowloris command (**slowloris 127.0.0.1**) the command told us it was attacking the server 127.0.0.1 with 150 socket, however no attacks were able to successfully penetrate the server due to the mitigation rules.
4. Finally, if we take a look at the B.Pcap file and filter DNS we notice all the traffic coming from 127.0.0.53 is only with DNS while the other traffic coming from 127.0.0.1 is from TCP, this shows that this server coming from 127.0.0.53 is vulnerable, while the server 127.0.0.1 has Dos Mitigation because no DOS attacks were successful under that server.








```
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